INCH-POUND

MIL-P-80085C
25 January 1991
SUPERSEDING
MIL-P-80085B
16 April 1985

MILITARY SPECIFICATION

POSITIONERS, WELDING, HEADSTOCK, TAILSTOCK, AND COMBINATION TYPES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of welding positioners, with the function of rotating workpieces for precision welding operations.

1.2 Classification. This specification covers positioners of the following types and sizes. The type and size to be furnished shall be specified (see 6.2).

Type I - Headstock type, powered.
Type II - Tailstock type, non-powered.
Type III - Headstock-tailstock, combination.

Size 1 - 2,500 pound capacity.
Size 2 - 3,000 pound capacity.
Size 3 - 4,000 pound capacity.
Size 4 - 5,000 pound capacity.
Size 4a - 6,000 pound capacity.
Size 5 - 7,500 pound capacity.
Size 6 - 10,000 pound capacity.
Size 7 - 12,500 pound capacity.
Size 8 - 15,000 pound capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3436

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

Size 9 - 20,000 pound capacity.
Size 10 - 24,000 pound capacity.
Size 11 - 30,000 pound capacity.
Size 12 - 37,000 pound capacity.
Size 13 - 50,000 pound capacity.
Size 14 - 60,000 pound capacity.
Size 14a - 75,000 pound capacity.
Size 15 - 90,000 pound capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by The Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, and Tongues. (DOD adopted)

ANSI B5.10 - Machine Tapers. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363).

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems. (DOD adopted)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete positioner shall be subject to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The positioner shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements specified herein. The positioner shall support and rotate workpieces as necessary for performing effective downhand and irregular type welding operations. The positioner shall include all components, parts, and features necessary to meet the performance requirements specified herein. Additional features which are not specified herein, that are a part of the manufacturer's current standard model, shall be included in the positioner furnished. The type I units shall have a self-locking device to prevent inadvertent rotation of the table. All type I and type II positioner units of the same capacity shall be designed with matching dimensions, except the type I units shall be designed with power and controls. The positioner design shall be such as to provide convenient, efficient, and safe operation. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The positioner shall be designed to operate satisfactorily in ambient temperature from 0 to 120 degrees Fahrenheit.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Dials. All dials used to indicate rotation motion shall be graduated in increments not larger than one degree. Dials that indicate inches per minute (IPM) shall be graduated in .01 inch per minute. Dials that indicate tilting degrees shall be graduated in one degree per revolution. Position indexing shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.2.3 Reclaimed materials. The positioners may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the positioner. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

3.2.4 Energy efficiency. The positioner and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.5 Controls. All mechanical, electrical and electronic operating controls shall be located convenient to the operator's work station(s).

3.2.6 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the positioner , whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910. Additional safety and health requirements shall be as specified (see 6.2).

3.2.7. Cover, guards, or other safety devices. Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial positioner supplied to the commercial market shall be provided for the point of operation and all other parts of the positioner that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration the procuring activity shall provide part design data to the supplier of the positioner (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the operation of the positioner. All guards shall allow access to the guarded parts.

3.2.8 Noise level. Unless otherwise specified (see 6.2), noise level emitted by the positioner when operating under a no load condition shall not exceed 84 db when measured on A scale of a type I or type II sound level meter at slow response.

3.2.9 Mercury restriction. The positioner shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.10 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the positioner.

3.2.11 Environmental protection. During the manufacture, operation, service, transportation, or storage of the positioner, and its components materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.12 Lubrication. An automatic lubricating system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.13 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or positioner.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

3.3 Construction. The positioner shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the positioner to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the positioner and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The positioner shall consist of not less than a frame and base assembly, table , horizontal spindle, drive system (type 1 units only), and all the necessary controls to comprise a complete positioner system.

3.4.1 Type I positioner. The type I positioner shall be provided with not less than a frame and base assembly with horizontal shaft or rotating spindle and a vertical face plate or worktable. The type I unit shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

power driven and equipped with self-locking worm or spur gears, (AC) or (DC) motor drive, or electronic drive system and all the necessary controls to comprise a complete positioner system.

3.4.1.1 Type II positioner. The type II positioner shall be equipped with not less than a frame and base assembly with a non-powered free-wheeling unit providing support to the workpiece.

3.4.1.2 Type III positioners. The type III positioner shall be equipped with one type I unit and a matching type II unit with provisions for in line mounting. The type III unit shall consist of a frame and base assembly, horizontal spindle housings and supporting face plates or worktables. Both units shall be mounted on a common surface or rails.

3.4.2 Frame and base assembly. The frame and base assembly shall be of steel weldment construction and reinforced to maintain alignment to minimize deflection under maximum capacity load conditions. The base shall be designed for floor or platform mounting. When optional sub-base frames are required, they shall be as specified (see 6.2).

3.4.2.1 Table. The table shall be of sufficient strength to maintain alignment for performance of downhand, irregular, and other welding operations at maximum capacity load conditions. Unless otherwise specified (see 6.2), the table work surface shall be machined flat to within $\pm$ 1/32 of an inch. Unless otherwise specified (see 6.2), the table shall be square, and be not less than the dimensions specified in table I. The table center height shall not be less than the height shown in table I.

3.4.2.2 Table T-slots. Unless otherwise specified (see 6.2), the table shall be furnished with four equally spaced T-slots . The T-slots shall conform to ANSI B5.1 and shall be of the self-cleaning type. The T-slots length shall be not less than 80 percent of the diagonal dimensions of the table. The table shall also be provided with four equally spaced pilot holes and four matching T-nuts for each table.

3.4.2.3 Horizontal spindle. Unless otherwise specified (see 6.2), a horizontal hollow spindle shall be furnished on all positioners. Spindle bearings shall be as required for continuous performance. Bearing tapers shall conform to ANSI B5.10, and shall be preloaded.

3.4.3 Drive system. The type I positioner shall be equipped with either an (AC) or (DC) electric motor drive or an electronic drive with self-locking worm or spur gears. The spindle torque shall be not less than that specified in table I. Unless otherwise specified (see 6.2), the table speed shall be either constant or variable speed and shall be not less than that specified in table I, and the variation from the control set point shall be not greater than 5 percent. The table rotation shall also be provided with variable speed ranges conforming to table I, with deviations of not more than 5 percent. When specified (see 6.2), a high speed electronic drive table shall be furnished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities.
Types I and II positioners.

Capacities	Positioner size																
	1	2	3	4	4a	5	6	7	8	9	10	11	12	13	14	14a	15
Unit load capacity (pounds)	2,500	3,000	4,000	5,000	6,000	7,500	10,000	12,500	15,000	20,000	24,000	30,000	37,000	50,000	60,000	75,000	90,000
Table size (inches per side square table)	30	30	40	48	48	48	48	54	54	54	54	54	54	54	54	54	54
Table center to floor (inches)	27	27	27	36	36	36	36	36	50	50	50	50	50	50	50	50	50
Table thickness (inches)	1	1	1-3/4	1-3/4	1-3/4	1-3/4	1-3/4	1-3/4	1-3/4	2	2	2	2	2	2-1/2	2-1/2	2-1/2
	Types I and III positioners only.																
Minimum ground current amperes	1,500	1,500	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	3,000	3,000	3,000	3,000	3,000	3,000
Table, constant speed (RPM)	1.13	1.10	1.00	1.00	.52	1.00	1.00	.50	.40	.40	.50	.50	.50	.50	.50	.50	.50
Table, variable speed range (RPM)																	
Minimum	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00
Maximum	1.00	1.00	2.70	2.00	.54	1.60	2.00	2.40	1.00	1.10	2.00	1.00	1.40	1.00	1.00	1.00	1.00
Spindle rotational torque (inches-lbs.)	15,000	36,000	36,000	48,000	72,000	90,000	120,000	150,000	240,000	240,000	288,000	360,000	444,000	600,000	800,000	800,000	1,080,000
	Type III positioners only.																
Load capacity between units (pounds)	5,000	6,000	8,000	10,000	12,000	15,000	20,000	25,000	30,000	40,000	48,000	60,000	74,000	100,000	120,000	150,000	180,000

All values shall be not less than those shown except where a range is shown, the values shall be not more than the minimum and not less than the maximum figures shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

3.4.4 Electrical requirements. The type I positioner shall be wired to conform to NFPA 79. Each positioner shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each type I positioner shall be designed to operate on 3-phase, 60-Hertz (Hz), 230/460-volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The positioner shall be initially wired as specified (see 6.2).

3.4.4.1 Motors. Motors shall meet the requirements of 3.4.3. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.4.2 Magnetic reversing starter and safety disconnect switch. Unless otherwise specified (see 6.2), the type I unit shall be furnished with a magnetic reversing starter and safety disconnect switch, both with overload protection. Speed drive controls shall be mounted in an enclosure conforming to NEMA ICS 1, style 1.

3.4.4.3 Headstock remote control. Unless otherwise specified (see 6.2), all Type I units shall be furnished with a remote control system in accordance with NFPA 79. The type I units shall be equipped with a remote pendant type hand operated control with a flexible multi-colored cable of not less than 15 feet. The pendant hand-operated control shall contain the following:

a. Tachometer with inches per minute (IPM) display.

b. Push button power "on" pilot light.

c. Push button on and off switch.

d. Push button forward and reverse switch.

e. An overload protection system.

f. Tachometer calibrated in revolutions per minute (RPM) for variable speed models.

3.4.4.4 Weld current grounding device. A weld current grounding device shall be furnished on all positioners . The capacity of the grounding device shall not be less than 1500 amperes direct current (DC) with suitable frame to ground cable attachment. The attachment device for securing the welding grounding cable to the positioner shall be permanently mounted in a position which shall prevent interference during full operational range of the positioner. The type of connection shall assure low resistance electrical conductivity. The welding current grounding device shall meet the requirements of NFPA 79.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the positioner shall meet the requirements of table I for the size positioner specified. Unit load and capacity for type I, II and III units is the weight of the workpiece that a unit will support with the center of gravity of the workpiece spaced 6 inches from the table face. Load capacity for the type III combination unit, is the weight of a workpiece plus welding fixtures that can be supported with the weight equally distributed between units. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2.), the physical size of the positioner shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Performance. The positioner shall be capable of supporting, positioning and rotating workpieces with performance capabilities of not less than those specified in table I. The positioner shall be capable of continuous operation and shall be capable of withstanding rough usage encountered by this type of equipment.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial positioner shall be furnished.

3.8 Optional equipment. Optional equipment specified in the following subparagraphs shall be furnished with each machine as specified (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or positioner for the type and size machine specified.

a. Round table
b. Pneumatic clamping
c. Foot operated controls
d. Interconnecting rails
e. Three jaw reversible chuck
f. Adapter plate

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.10 Marking on charts, and plates. All words on the machine including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

3.10.1 Lubrication plate or chart. When specified (see 6.2) a lubrication plate or chart shall be attached to each positioner. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each positioner. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below:

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Work load capacity (pounds)
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.11 Workmanship. Workmanship of the positioner and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The positioner shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each positioner prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.4. The positioner shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on each positioner to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the positioner is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The positioner shall pass the examination and all tests to be accepted.

4.7 Examination. Each positioner shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 degrees to 104 degrees Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. Each positioner shall be operated at no load for not less than 30 minutes. The positioner shall be connected to a power source and cycled through each axis of its entire range, addressing all feeds, speeds, direction of rotation and standard control functions specified in this specification. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

4.8.3 Performance test. The positioner shall be tested and inspected to determine compliance with the requirements of this specification. The type I, type II, and type III combination positioners shall be placed on a firm and level surface, or a track or rail system. A capacity load specified in table I for the size procured, shall be mounted on the face plate or table with the center of gravity and overhang off-center and located 6 inches from the table or face plate . The positioner unit and its component parts, and accessories shall be connected to a power source and operated through all feeds, speeds, increments, and all other operations over the entire range of the positioner. During the performance, each operational sequence shall be inspected, tested, and verify the performance requirements specified herein. During the performance period, proper operation of all controls, motors, and gears shall be tested and verified and be complete without making any adjustments to or performing any maintenance on the positioner or its component parts. The positioner table shall rotate freely with no evidence of galling or binding.

4.8.3.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8.3 shall be as specified and fully described (see 6.2).

4.8.4 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirement for packaging shall be in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The positioner is used for positioning and rotating weldments with accuracy, efficiency, and repeatability for effective downhand and irregular welding operations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

f. Additional safety and health requirements, if required (see 3.2.6).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.7).

h. Specify noise level requirements, if different (see 3.2.8).

i. Specify surface finish requirements, if different (see 3.3.3).

j. Painting, if different (see 3.3.5.).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

l. Optional frame, if required (see 3.4.2).

m. Table surface tolerance , if different (see 3.4.2.1).

n. Table shape and dimensions, if different (see 3.4.2.1).

o. Table T-slots, if different (see 3.4.2.2).

p. Hollow spindle, if different (see 3.4.2.3).

q. Table speed, if different (see 3.4.3).

r. High speed table, if required (see 3.4.3).

s. Voltage requirements, if different (see 3.4.4).

t. Initial voltage, as specified (see 3.4.4).

u. Motors, if different (see 3.4.4.1).

v. Magnetic reversing starter and safety switch , if required (see 3.4.4.2).

w. Control system, if different (see 3.4.4.3).

x. Size and capacity, if different (see 3.5).

y. Specify physical size limitation, if required (see 3..5.1).

z. Additional performance features, if required (see 3.6.1.).

aa. Specify and fully describe optional and accessory equipment, if required (see 3.8).

bb. Specify applicable repair parts, if required (see 3.9).

cc. Lubrication plate, if required (see 3.10.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

dd. Lubrication plate information, if different (see 3.10.1).

ee. Nameplate, if different (see 3.10.2).

ff. Inspection conditions, if different (see 4.3).

gg. First article inspection, if different (see 4.4).

hh. Quality conformance inspection, if different (4.5).

ii. Acceptance test, if different (see 4.6).

jj. Additional performance tests, if required (see 4.8.3.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from previous of this specification are as following:

MIL-P-80085B	MIL-P-80085C
	Size 4a (new size)
	Size 14a (new size)

6.8 Changes from previous issues. Marginal notations are not used in this revision to idenyify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual positioners an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

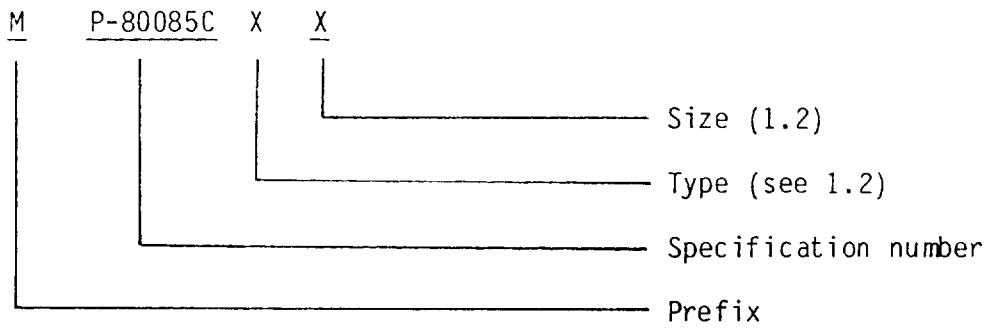
6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PIN to be used for positioners acquired to this specification are created as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.



6.14 Subject term Key word listing.

Combination downhand and irregular welding surfaces
Headstock remote control
Headstock unit
Magnetic reversing starter
Positioner type rotation table
Tailstock unit

Custodians:

Army - AR
Navy - SH
Air Force - 99
DLA - GS

Preparing activity:

DLA - IP

(Project 3436-0043)

Review activities:

Army - CR, ER
Navy - AS, OS
Air Force - 84
DLA - GS

User activities:

Navy - MC
Air Force - 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-80085C	2. DOCUMENT DATE (YYMMDD) 91 01 25

3. DOCUMENT TITLE
POSITIONERS, WELDING, HEADSTOCK, TAILSTOCK, AND COMBINATION TYPES

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	b. TELEPHONE (Include Area Code) (1) Commercial: 901-775-6699 (2) AUTOVON: 683-6699
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-04-03T04:37
Check the source to verify that this is the current version before use.